A

Compound-class distribution across generated sets

	intermetallic	oxide	metal	chalcogenide	salt/ionic compound	halide	boride	pnictide	silicide	oxyhalide	Other
MatterGen Base	48.8	14.8	0.9	1.7	2.1	4.2	2.1	3.8	2.5	4.6	14.5
MatterGen FT	55.0	0.0	1.6	0.1	17.4	0.0	2.5	5.4	12.8	0.0	5.1
Con-CDVAE Base	14.4	21.0	1.9	14.4	1.6	14.0	0.3	4.3	0.7	13.5	13.8
Con-CDVAE FT	18.6	0.0	4.8	1.8	14.2	0.0	27.7	9.5	6.3	0.0	16.9
CrystalFormer Base	28.9	11.7	24.5	8.8	2.4	4.2	0.7	3.3	2.0	0.4	13.1
CrystaLLM Base	29.6	45.9	0.0	0.0	0.0	12.2	8.2	0.0	4.1	0.0	0.0
Crys-JEPA Base	30.3	16.3	1.7	13.2	4.7	11.3	2.2	4.0	5.0	3.1	8.3
Crys-JEPA Ref.	76.1	0.0	20.2	0.3	0.0	0.1	0.4	0.9	0.1	0.0	1.9

Share within generated set (%)

B

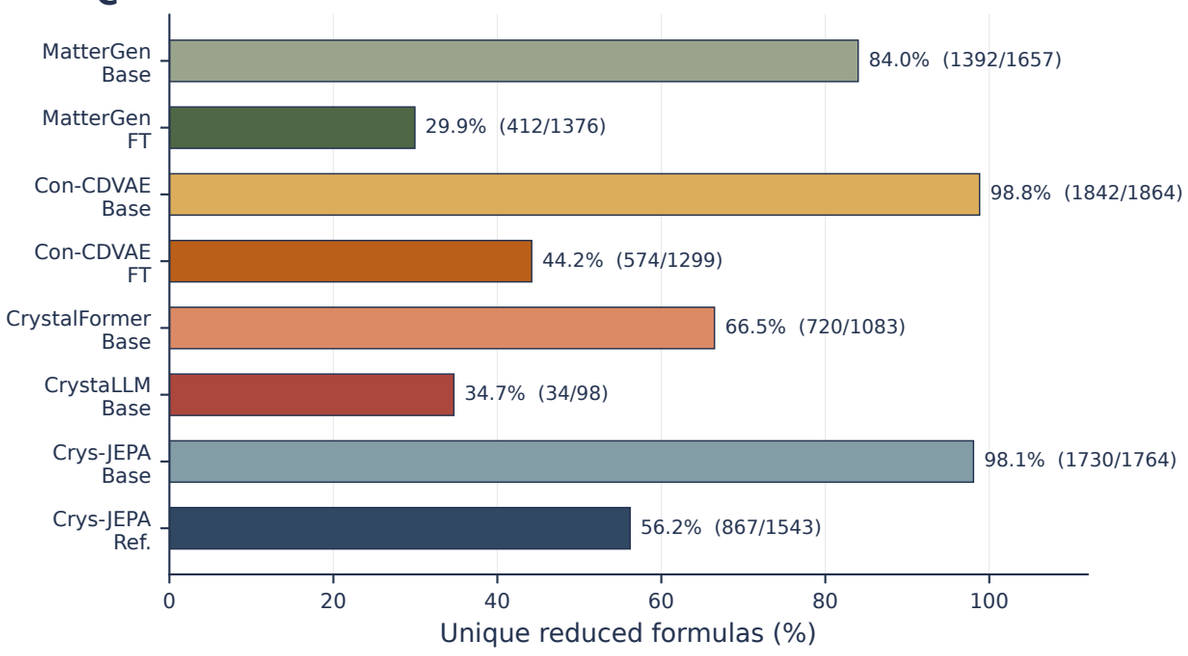
Fine-tuning and refinement shifts in compound classes

	intermetallic	oxide	metal	chalcogenide	salt/ionic compound	halide	boride	pnictide	silicide	oxyhalide	Other
MatterGen Fine-tuned − Base	+6.2	-14.8	+0.7	-1.5	+15.3	-4.2	+0.4	+1.6	+10.3	-4.6	-9.5
Con-CDVAE Fine-tuned − Base	+4.2	-21.0	+2.9	-12.7	+12.6	-14.0	+27.4	+5.3	+5.6	-13.5	+3.1
Crys-JEPA Refined − Base	+45.8	-16.3	+18.5	-12.8	-4.7	-11.2	-1.8	-3.1	-4.9	-3.1	-6.4

Change from baseline (percentage points)

C

Unique reduced formulas



D

Unique formula–space-group pairs

